

Smart glasses that listen for machine faults.

A self-funded hardware pivot that became a benchmarked audio ML system.

WHY I BUILT IT

About six months ago, my boxing coach came to the gym wearing Ray-Ban Meta glasses. Seeing someone I knew use smart glasses made the technology feel real. I wanted to build for that future, so I completed coding bounties and programming work for my dad, saved the money, and bought Even Realities G2 glasses myself. Applied Resonance grew from one question: could smart glasses give a technician another way to notice a machine problem before it becomes obvious?

THE IDEA



Learn a healthy sound. Flag the change.
Explain it in two lines.

HOW APPLIED RESONANCE WORKS

01

LISTEN

G2 mic or audio

02

LEARN

30 s baseline

03

COMPARE

3 s windows

04

SCORE

PANNs + kNN

05

ALERT

two-line HUD

Current implementation: local scoring engine, no cloud inference required



Bearing-like anomaly | 98%

high-band energy rising | tap to log

TWO-LINE G2 HUD CONTRACT

WHAT MAKES IT DIFFERENT

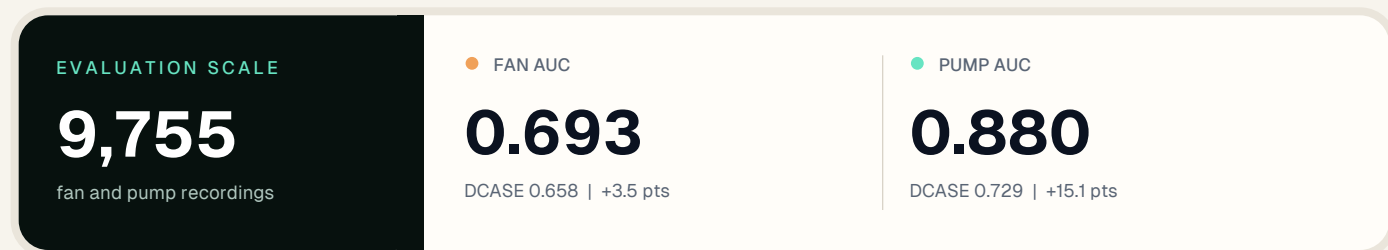
Most monitoring systems attach a sensor to one machine. Applied Resonance explores a different model: the microphone moves with the technician, while the display stays hands-free. The same pair of glasses can move between machines, surface an anomaly, and save evidence for later review.

BUILT WITH

Python / PyTorch
TypeScript / Even Hub
PANNs / kNN / ONNX

What I proved, and what I still need to prove.

The benchmark is proven. The real G2 microphone is the next gate.



DCASE is an international research challenge for detecting and classifying sounds. Its 2020 Task 2 autoencoder is the official reference model for machine anomaly detection. Applied Resonance was fit only on healthy machine clips, then evaluated on held-out healthy and faulty clips. AUC is a ranking metric where 0.5 is chance and 1.0 is perfect.

THE FAILURE I DID NOT HIDE

A fan-versus-pump classifier reached 1.000 training accuracy, then dropped to 0.577 on unseen machine IDs. I removed it from the live path instead of presenting the training score. The anomaly detector uses a separate per-machine design and was unaffected.

HOW I BUILT IT

This was my first Claude Code build with Fable 5. I split it into 22 independently testable tasks, wrote acceptance criteria, and checked each result. Faster generation made verification more important, not less.

VALIDATED TODAY

- Full MIMII fan and pump evaluation
- Deterministic double-pass across all reported metrics
- Live streaming, evidence, and labeling engine
- Even Hub simulator and real-engine integration

NEXT G2 HARDWARE GATE

- Confirm continuous PCM capture from the real glasses
- Verify sample rate with a known 1 kHz tone
- Re-record the pump test through the G2 microphone
- Measure background operation and battery cost

The part I am proudest of is that I was willing to find out where it did not work.

Full 63-page build transcript included in the application folder | Source: engine/REPORT_FULL.md